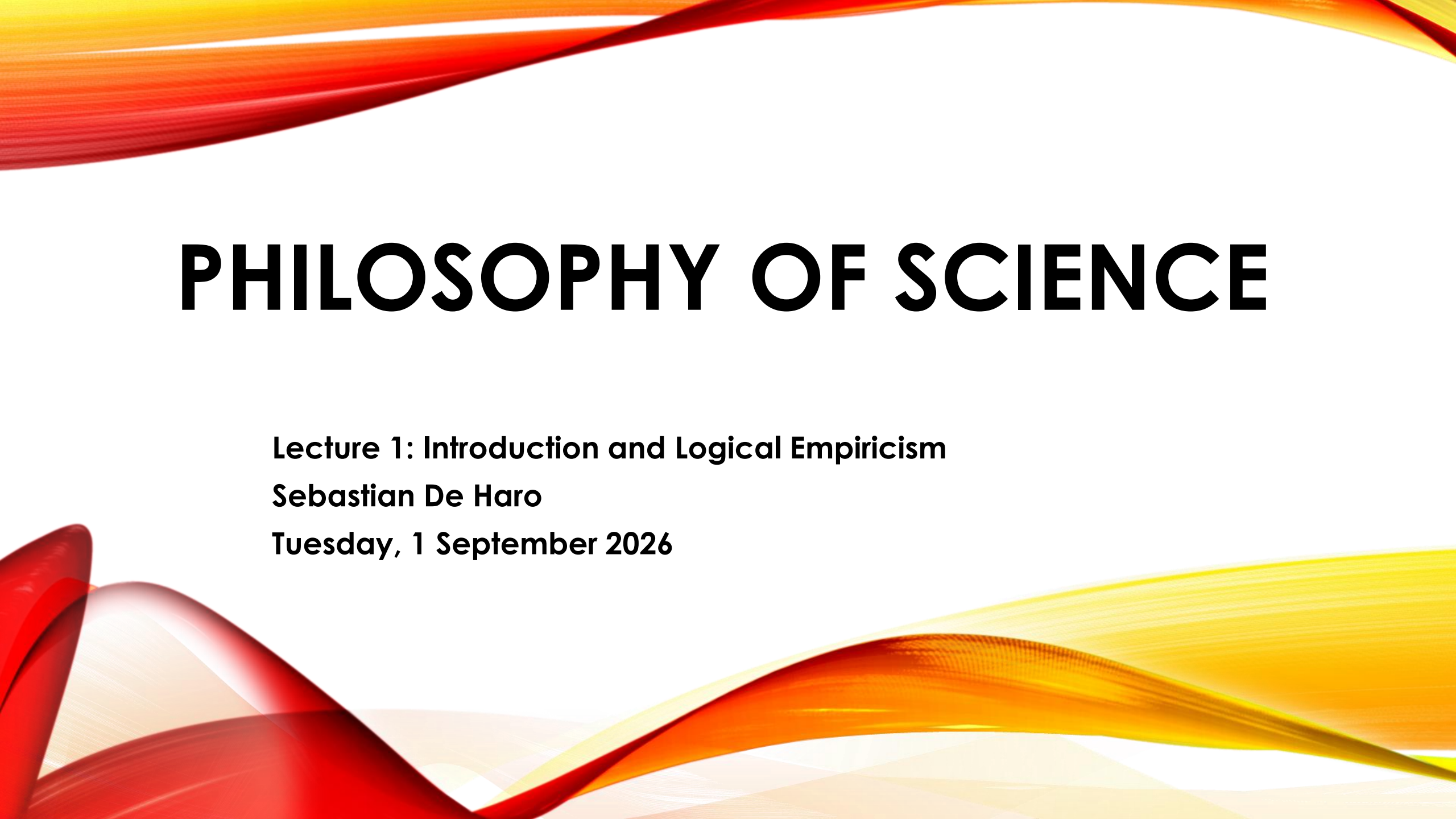




# **PHILOSOPHY OF SCIENCE**

**Lecture 1: Introduction and Logical Empiricism**

**Sebastian De Haro**

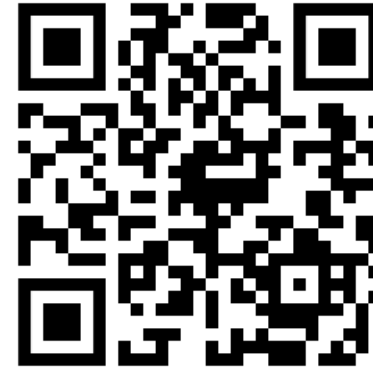
**Tuesday, 1 September 2026**

An abstract, flowing, ribbon-like shape in shades of red and orange, positioned on the left side of the slide.

# OUTLINE

- Short introduction
  - (A) Philosophy of science: brief history of the field and content of the course
  - (B) Practical matters
  - (C) Rudolf Carnap: *meaning and verification, and against metaphysics*
  - (D) Logical empiricism: *What is a scientific theory?*

# LECTURER

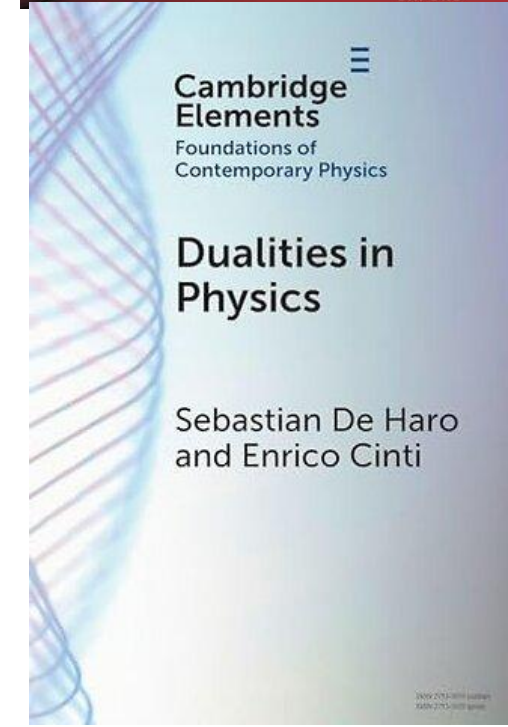
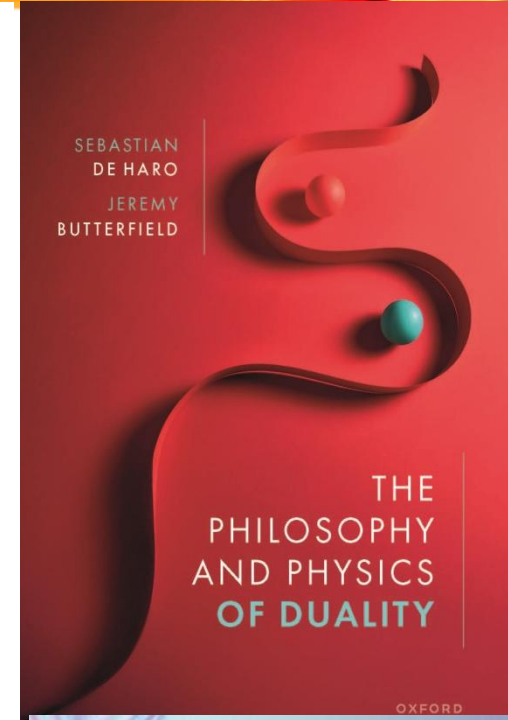


- Institute for Logic, Language and Computation
- Institute of Physics
- QuSoft
- Shaping interfaces between science and the public, IAS

Work in:

- Philosophy of science and technology
- Philosophy of physics
- Theoretical and mathematical physics

<http://www.uva.nl/en/profile/s.deharoolle>





# (A) PHILOSOPHY OF SCIENCE IN BRIEF

Philosophy, modern physics and logic  
The Vienna Circle



# PHILOSOPHY



Plato / Aristotle  
427–347 BC

**Plato: rationalism**  
**Aristotle: empiricism**  
Syllogistic logic



David Hume  
1711-1776

**Empiricism**



Immanuel Kant  
1724–1804

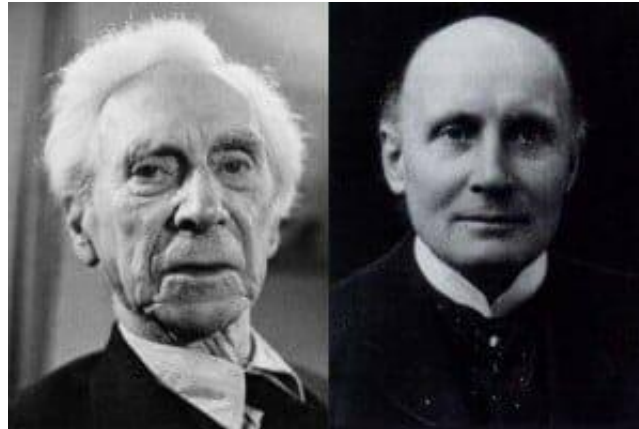
**Synthesis rationalism/empiricism**

# PHILOSOPHY AND MODERN LOGIC



Gottlob Frege  
1848–1925

First-order logic  
Philosophy of language



Alfred North Whitehead 1861–1947  
Bertrand Russell 1872–1970

Deriving mathematics from logic

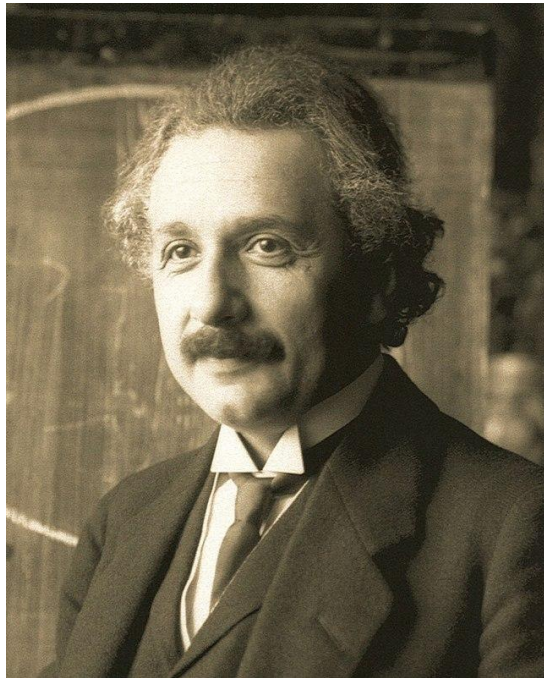


Ludwig Wittgenstein  
1889–1951

Relation language-reality  
Meaning as use



# PHILOSOPHY AND MODERN PHYSICS



Special theory of relativity 1905  
General theory of relativity 1915



Niels Bohr, Werner Heisenberg, Erwin Schrödinger  
Quantum mechanics 1900-1927

# KANT'S LEGACY AND ANALYTIC PHILOSOPHY

- 20th century **divide analytic-continental** philosophy:
- *The fundamental cleavage in 20th-century philosophy derives from just this dispute over the **priority of logic or fundamental philosophical insight** in grounding objective knowledge (Richardson, 1996)*

Rudolf Carnap  
1891-1970  
Jena



Martin Heidegger  
1889-1976  
Freiburg



# THE VIENNA CIRCLE

1924-1938



**Rudolf Carnap**



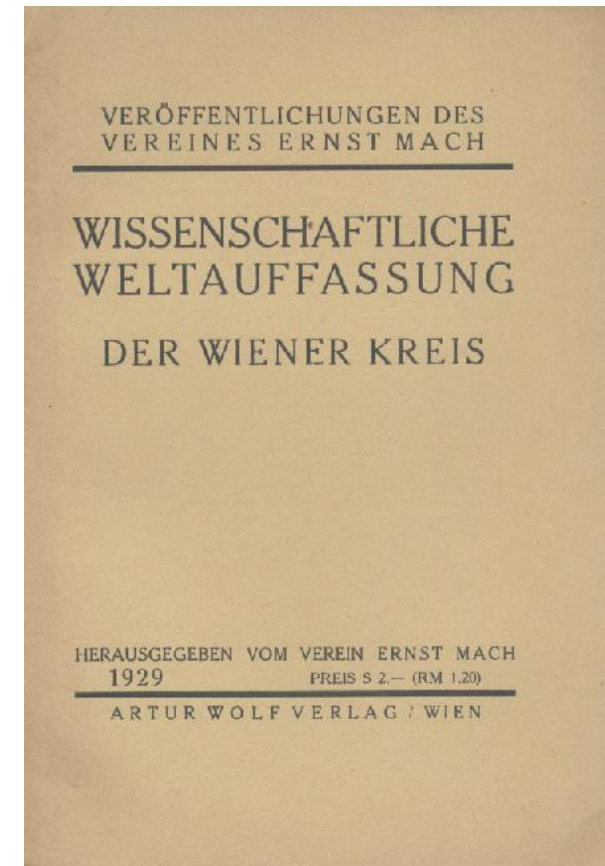
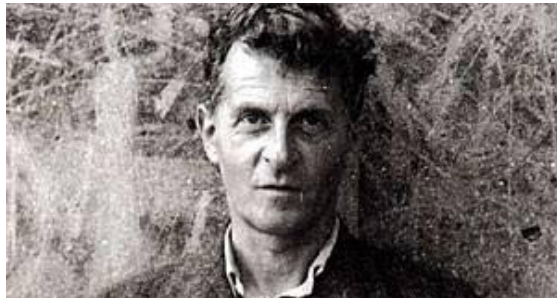
**Moritz Schlick**



**Otto Neurath**



**Kurt Gödel**



# END OF VIENNA CIRCLE

- Moritz Schlick **murdered** in 1936 by former student Johann Nelböck
- **Annexation** Austria by Nazi Germany 1938
  - Rudolf Carnap through Prague → Chicago 1936
  - Otto Neurath through the Hague → London 1940
  - Popper → Cambridge 1937
  - Gödel → Princeton 1938
- Early period (Vienna): *logical positivism*
- Later called **logical empiricism**



# TIME LINE 20TH CENTURY



## First half 20th century:

- Normative/regulative ideal of science
- Justification of scientific knowledge

## Second half 20th century:

- How does science actually work?
- What do scientists do?



# LATE 20TH, EARLY 21ST CENTURIES

- **Deepening** science and society studies, philosophy of technology, and connection with scientific practice
- **Recent revival of logical empiricist views**, corrected and enriched by other ideas
- Topics developed against the backdrop of logical empiricism:
  - Models
  - Explanation
  - Scientific understanding

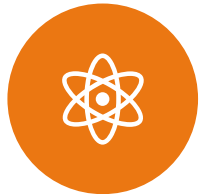


Nancy Cartwright



Margaret Morrison

# SIX CENTRAL DEBATES



What is science?  
Logical empiricism



Kuhn on scientific  
progress



Duhem-Quine  
under-  
determination thesis



Explanation and  
understanding



Scientific realism  
(2x)



Laws of nature



# PHILOSOPHICAL WORDS

- **Metaphysics:** the study of the nature of things/being
  - **Includes:** space/time, causation, freedom/determinism, mental/physical
  - **Ontology:** subfield about *being as such* (what things there are)
- **Epistemology:** knowledge/understanding
- **Semantics:** study of meanings of words, including in formal/mathematical languages
  - Contrast with **syntax:** formal rules of the language



## Metaphysical positions

### Realism about *objects*

- Objects exist independently of our minds and knowledge of them
- Our knowledge is constrained by, and aims to represent, such objects
- [Aristotle, Descartes, Locke]

### Realism about *ideas*

- Universals/ideas/general concepts correspond to entities that exist independently of the mind
- Such entities do not depend on our knowledge of them
- [Plato; in a different sense, Aristotle]

### Idealism

- Objects, or objects as they are known, are in some significant sense dependent on mind, experience, or conceptual activity
- The precise claim differs between authors
- [Berkeley, Hegel, Schopenhauer]

### Nominalism

- There are only individual things; general terms do not correspond to mind-independent universals
- Generality is explained through names, conventions, or mental concepts
- Nominalism is compatible with realism about physical objects
- [Ockham]

## Epistemological Positions

### Rationalism

- Reason can provide substantive knowledge that is independent of sense experience.
- Mathematics, clear and distinct ideas, axioms, and deduction are central sources and methods of knowledge.
- Rationalists seek certainty and necessity in at least some domains of knowledge
- [Plato, Descartes, Spinoza, Leibniz]

### Empiricism

- Experience and sense perception are the fundamental source of substantive knowledge about the world.
- Induction is central to empirical inquiry.
- Empirical knowledge is generally contingent and fallible, rather than certain
- Empiricism is compatible with idealism.
- [Aristotle, Locke, Berkeley, Hume, van Fraassen]

### Kant

- Some judgements are synthetic a priori.
  - Synthetic: they extend our knowledge.
  - A priori: their justification does not derive from particular experiences.
- Space, time, and the categories are conditions under which objects can be experienced by us.
- Seeks to combine empiricism about the content of knowledge with rationalism about its a priori conditions.

(B) PRACTICAL MATTERS



# TEACHING TEAM

- Sebastian
- Enrico: co-lecturer
- Luka: tutorials
- Matteo: tutorials



# PREPARATION AND LECTURES

- **Preparation:** read texts before class
  - Two/three texts each week
  - Chapter introductions and commentary are strongly recommended!
- **Examinable material:** not just texts in syllabus
  - **Lectures** always additional explanation/background/material: relevant for **exam**
  - See **slides** and additional materials on Canvas
- **Experiment with flipped classroom: lectures 3 and 5 (15 and 29 September)**
  - Lecture on 15 September will include guidance on use of AI when studying philosophy



# ASSIGNMENTS

## **Paper (30%)**

- Allows you to go deeper into one of the topics of the course or a closely related topic
- Guidance by Luka:
  - Writing workshop where you discuss your project idea
  - Project proposal
  - Draft version and final project

## **Presentation (20%)**

- Group presentation about your project during the last tutorial
- See guidelines on Canvas
- More information on Thursday during the tutorial

## **Exam (50%)**

- On Monday 19 October (check Datanose for time and location)



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# (C) RUDOLF CARNAP

- (1) Meaning and verification
- (2) Against metaphysics

# RUDOLF CARNAP

- Frege, Russell, Wittgenstein: **tautological** nature of logic
  - Logic adds **no new content**
- **Vienna Circle** takes it as one of their principles: meaningful statements either **empirically verifiable** or **analytic (logic/math)**
- Early Carnap: function of philosophy is **logical/linguistic analysis**
  - **Syntactic**: look at the forms of reasoning, rule-following
  - **Semantics (meanings)** considered part of **psychology**
  - **Formal semantics** only late 1930s / 40s (Alfred Tarski)
- Hence Carnap's definition of the **meaning of a word** as:
  - (1) **Deducibility** of/from elementary sentence S
  - (2) **Conditions for truth** of S
  - (3) **Verifiability** of S
  - (4) **Meaning** of S





# RUDOLF CARNAP

- Frege, Russell, Wittgenstein: **tautological** nature of logic
  - Logic adds **no new content**
- **Vienna Circle** takes it as one of their principles: meaningful statements either **empirically verifiable** or **logically necessary**
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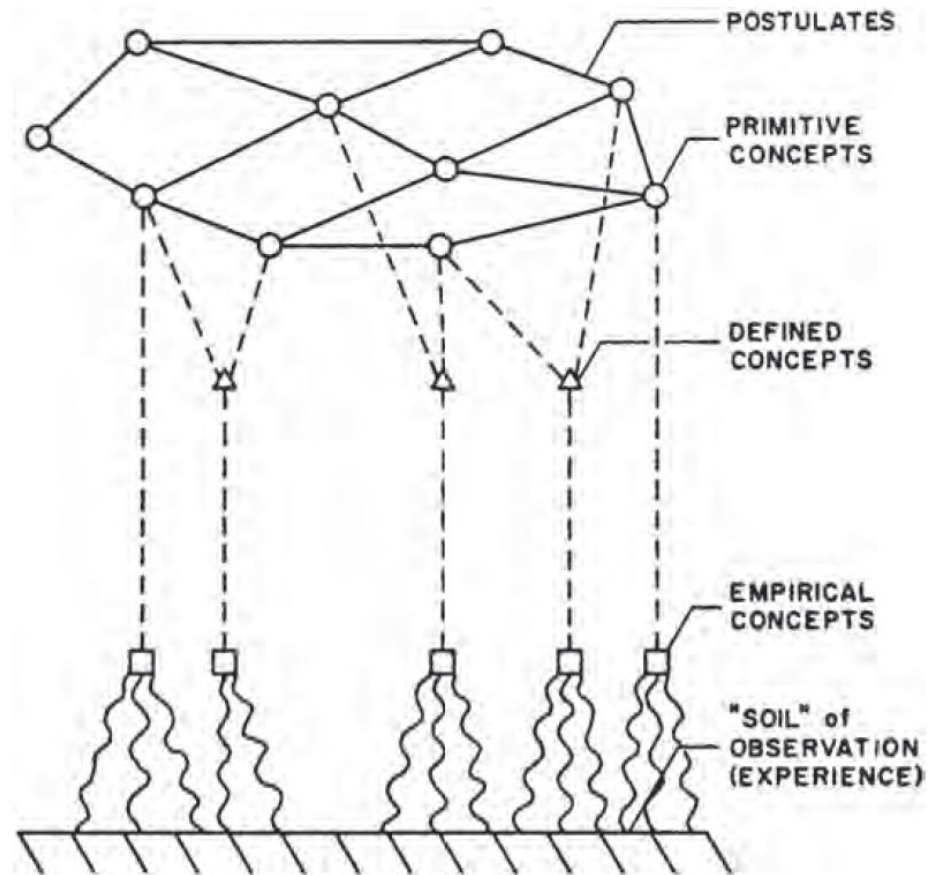


**Questions:** Are deduction/truth/verification/meaning the same concepts? How can these four definitions give the same concept of meaning?

# (1) MEANING AND VERIFICATION

- This is because of the way Carnap thinks about a **scientific theory**
- Scientific theory = set of sentences deduced from **observation sentences**
  - Truth of observation sentences is verified by **observation**
- A word is meaningful if the sentences in which it may occur are **reducible** to observation sentences (' $x$  has segmented body')
- **Elementary sentence** S: ' $x$  is an arthropod'
  - **Meaning** of S: ' $x$  is animal', ' $x$  has segmented body', ' $x$  has jointed legs'
- Arthropod' = 'animal with exoskeleton, segmented body and jointed appendages'
  - **Exoskeleton**: hard outer shell made of chitin.
  - **Segmented body**: bodies are divided into segments.
  - **Jointed appendages**: limbs with joints, like legs or antennae.





## (2) AGAINST METAPHYSICS

- In metaphysics... everyday language has led to confusions of types which, unlike those in everyday language, are no longer **translatable into logically correct form**. Pseudo-statements of this kind are encountered in especially large quantity, e.g. in the writings of Hegel and Heidegger... (e.g. predicates which should be applied to objects of a certain sort are instead applied to predicates of these objects or to "being" or to "existence").  
(Elimination, 6)
- Heidegger's 1929 Freiburg inaugural lecture, *What is Metaphysics?*
  - 'Das Nichts selbst nichtet': The Nothing itself noths (nihilates, nothings)
  - Modern logic: 'nothing' is neither noun, nor verb:  $\sim \exists x Px$
- Descartes 'I think therefore I am'
  - Correct form of 'something is' = 'exists' is  $\exists x$  (existence is not an activity)
  - 'I think' means 'there is something that thinks',  $\exists x Px$ .
  - A metaphysical notion of 'ego' or 'I' is not needed: removes the subjective component of the statement.

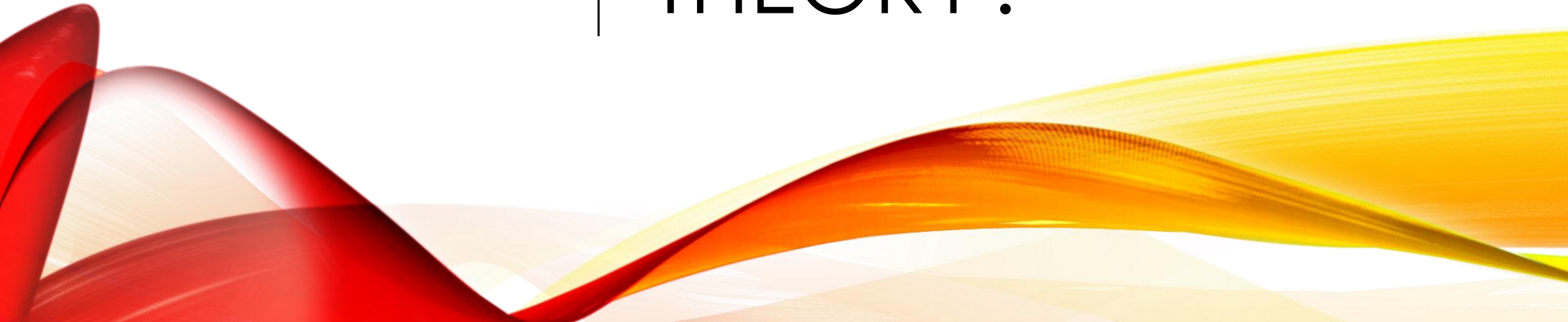


# CARNAP IN 1966 AND SCIENTIFIC THEORIES

- When I was young and part of the Vienna Circle, some of my early publications were written as a reaction to the philosophical climate of German idealism. As a consequence, these publications ... were **filled with prohibitory statements**. These prohibitions must be understood in reference to the **historical situation** in which we found ourselves. Today, especially in the United States, we seldom make such prohibitions.
- 'The move to semantics blunts the force of Carnap's diagnosis of metaphysics as confusion' (Richardson, 1996)
- Insists that philosophical thinking **must be logical/rigorous**
  - **Logic and scientific way of thinking:** starting points of philosophy
- **Important question:** what is a **scientific theory**?

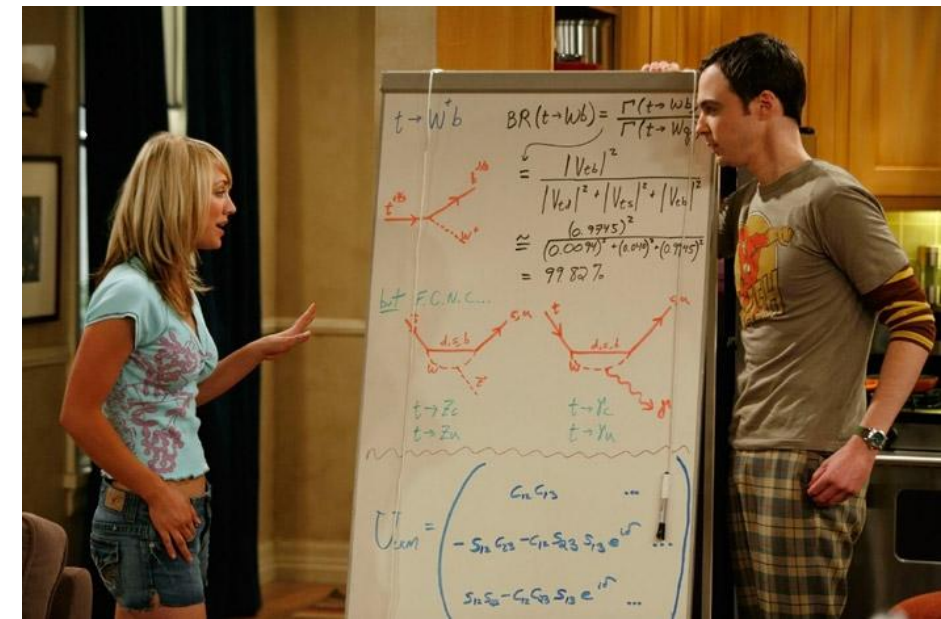
- (1) The syntactic conception
- (2) The semantic conception

# (D) LOGICAL EMPIRICISM: WHAT IS A SCIENTIFIC THEORY?



# (1) THE SYNTACTIC CONCEPTION

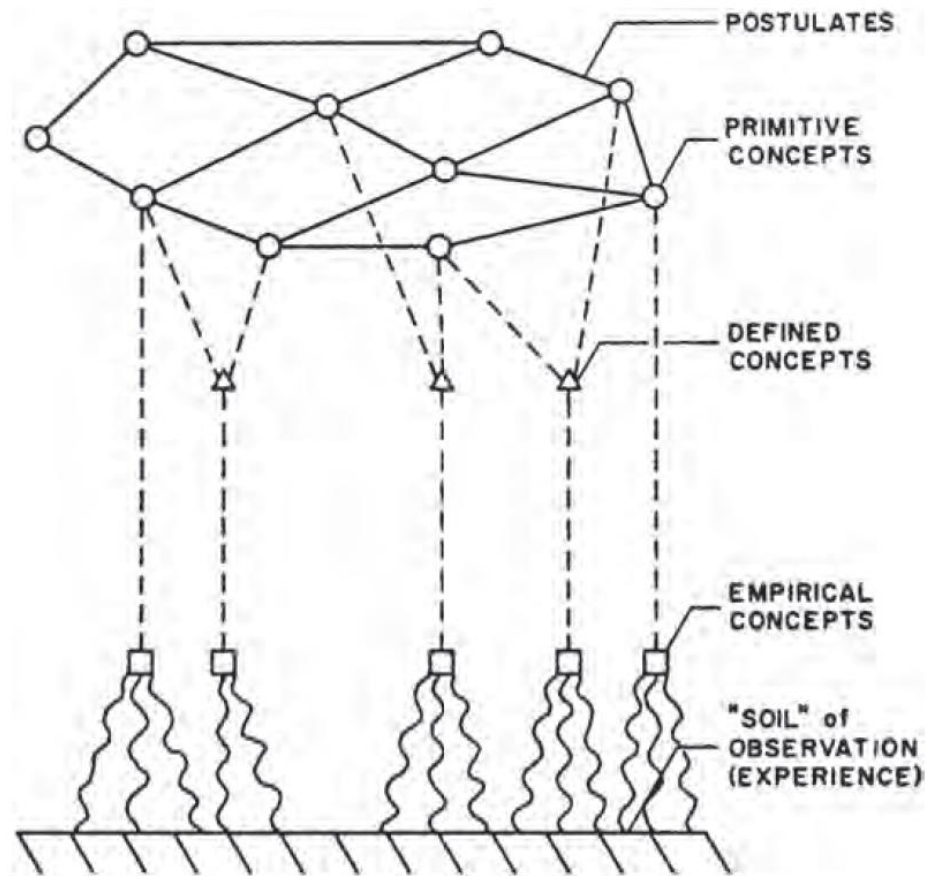
- **Syntactic conception of theory: collection of sentences** formulated in an appropriate system of **formal logic** such that:
  - Sentences are either **axioms** or deduced from axioms (interpreted as laws)
  - **Correspondence rules** relate theoretical terms to observational terms
- **Theory-observation distinction:**
  - **Theoretical terms** (e.g. 'atom', 'electron') not directly observable
  - Yet meaningful, because linked to empirical world (interpreted) through *correspondence rules*
- **Received View** of Theories (Putnam)



# THE SYNTACTIC CONCEPTION

- 'Syntactic', because it represents a scientific theory as a formal system: as a **set of sentences** in a (formal, or mathematical) language, i.e. with a vocabulary and rules of syntax (including deduction).
  - Distinction: observational/theoretical terms.
  - Observation sentences only contain observational terms in the theory's vocabulary.
- 1. **Axioms:** theoretical starting-points from which the theory is derived (basic laws).
- 2. **Other sentences are derived** from the axioms.
- 3. **Correspondence rules** between **theoretical (non-observational) and observational terms**.
- **Normative ideal:** it aimed to provide a clear and rigorous standard for evaluating the quality and progress of scientific knowledge.





# THE SYNTACTIC CONCEPTION

# EXAMPLE: EMPIRICAL GAS LAWS AND KINETIC THEORY

## Axioms of kinetic theory:

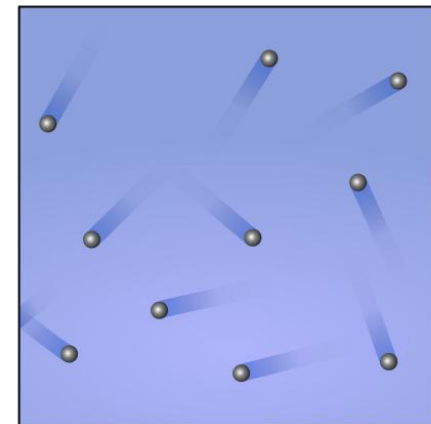
- Gases are systems of molecules in motion
- Energy and momentum are conserved
- Molecules are elastic, behave according to laws of Newtonian mechanics

## Correspondence rules (observation-theory link):

- Pressure = mean force with which molecules hit the walls of container
- Temperature = proportional to mean kinetic energy of molecules:  $kT = \frac{2}{3} \left\langle \frac{1}{2} mv^2 \right\rangle$

Kinetic theory yields:  $PV = \frac{2}{3} N \left\langle \frac{1}{2} mv^2 \right\rangle$

→ Empirical law:  $PV = NkT$  (Boyle-Charles ideal gas law)



# FINAL VERSION OF THE RECEIVED VIEW

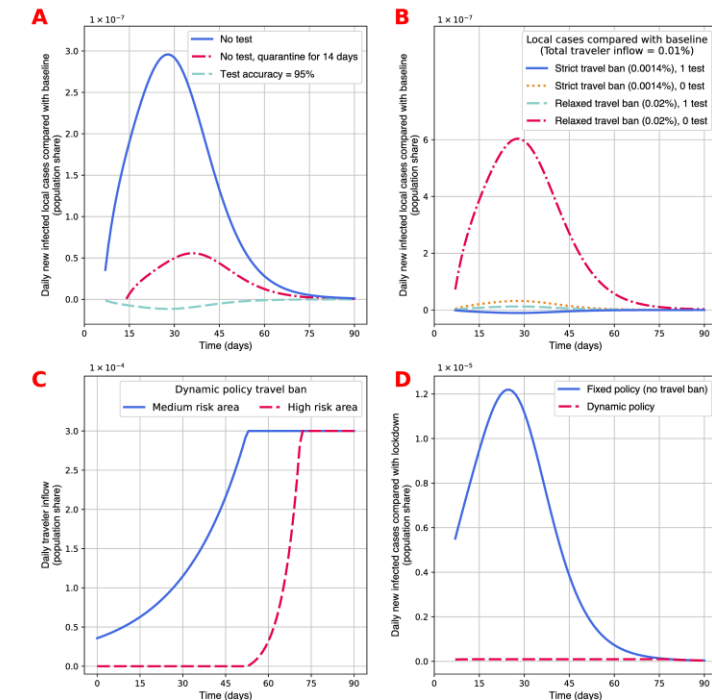
- Carnap was an eminently *open-minded thinker*:
- 1930s Carnap meets the logicians **Gödel and Tarski**:
  - Drops strict verificationism: meaning is not just verification.
  - Accepts the **interpretation of theoretical terms** i.e. a full **semantics**.
- Upholds his rejection of metaphysics, but this rejection loses force, since he **cannot avoid ontology**.
- The notion of **correspondence rules** is **weakened**:
  - Allows for **partial interpretation** of theoretical terms.
  - This is much closer to a realist view of science.
- Revival of **analytic metaphysics**: W.V.O. Quine, 1948 onwards.



## (2) THE SEMANTIC CONCEPTION

### Criticisms of syntactic view / motivation for semantic:

- **Axiomatization:** many scientific theories not in axiomatized form, perhaps will never be axiomatized
- **Language dependence:** any given theory may have several (equivalent) syntactic formulations, and so it seems wrong to identify a theory with one of its formulations
- **“A scientific theory is a collection of *models*”**
- **Models:** scientists use models:
  1. **Iconic / representational:** concrete, capture content of theories well  
**Structurally similar** (isomorphic) to what they model
  2. **Mathematical / logical model:** structure that makes the (sentences of) theory **true**





# EXAMPLE: KINETIC THEORY IN THE SEMANTIC VIEW

- On the semantic conception, kinetic theory is not primarily a set of sentences, but a class of **mathematical models representing possible gases**.
- **Model** a gas of  $N$  molecules in a region  $\mathcal{C} \subseteq \mathbb{R}^3$  of volume  $V$  by microstates in phase space,  $\Gamma_N = \mathcal{C}^N \times \mathbb{R}^{3N} \subseteq \mathbb{R}^{6N}$ , and laws of motion/collisions.
- Statistical mechanics then connects these micro-models to thermodynamics:
  - $V$  is fixed by the container,
  - $P$  is determined by average momentum transfer to the walls,
  - $T$  determined by the equilibrium distribution of molecular energies.
  - Under the idealising assumptions of kinetic theory, this yields:  $PV = NkT$ .
- Thus, on the semantic view, kinetic theory is identified with a **class of mathematical structures** that represent gases under specified conditions.
- The syntactic view emphasises axioms, derivation, and correspondence rules; the semantic view emphasises how mathematical structures apply to physical systems.

# THE SEMANTIC CONCEPTION

- “A scientific theory is a collection of *models*”
- **Models are structures:** defined in set theory: e.g. vector space.
- **Recent lively debate** about which of the two conceptions is superior:
  - Halvorson 2012, ‘What Scientific Theories Could Not Be’; 2013, ‘The Semantic View, If Plausible, is Syntactic’
  - Replies by Glymour (2013), van Fraassen (2014)
- **Consensus?** Lutz (2017): ‘the syntax-semantics debate really [does] not capture any significant differences’
- Frigg (2022: p. 69):
  - **Syntactic view** too strict: requires **models**, and natural language
  - **Semantic requires a language**
  - The two conceptions **collapse into each other!**



# CONCLUSION

- (A) Philosophy of science as **mature and dynamic** discipline
- (B) (Practical matters)
- (C) Rudolf Carnap on philosophy: “stay close to science”
  - (1) **Empirical verification**
  - (2) **Logical necessity** (analyticity)
- (D) Logical empiricist view of **scientific theories**
  - (1) **Syntactic conception** as **normative**
  - (2) **Historical dynamics** of scientific theories: Kuhn’s critique
  - (3) **Semantic conception**: the distinction collapses?

# FURTHER READING

- Excellent general resource: **Stanford Encyclopedia of Philosophy**
  - Logical Empiricism: <https://plato.stanford.edu/entries/logical-empiricism>
  - Vienna Circle: <https://plato.stanford.edu/entries/vienna-circle>

## More specialised texts:

- R.N. Giere and A.W. Richardson (Eds.) (1996). *Origins of Logical Empiricism*. In: *Minnesota Studies in the Philosophy of Science*, Volume XVI:
  - A.W. Richardson, *Introduction: Origins of Logical Empiricism* (**Canvas**)
  - M. Friedman, *Overcoming Metaphysics: Carnap and Heidegger* (**Canvas**)
  - P. Galison, *Constructing Modernism: Cultural Location of Aufbau* (**Canvas**)
- F. Suppe (1974). *The Structure of Scientific Theories* (**locus classicus**)
- R. Frigg (2022). *Models and Theories* (**state of the art**)